Real-World Scale Drawings

Teacher's Edition

The Period

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must Read a scale such as ¼ in : 1 ft or 1 : 50 as a rate, find real lengths and drawing lengths from it in both directions, and find a real area from the real dimensions. (MA.7.GR.1.5, clarifications 1 and 3; B1G-M's wall example.)

Must not Converting the real area into another system (5.08–09); drawing a plan to scale with a ruler (not a FAST shape — the audit's ruler items are out); a scale factor between two drawings of one object (5.01).

Target I can use a scale such as ¼ in : 1 ft to find real lengths and areas from a drawing, and drawing lengths from real ones.

MTR **MTR.1.1** — Example 2 and its Your Turn: the scale is not printed, and the room has to make it from one pair before the question can start · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.7.1** — every board is a plan, a map or a model: the scale is a real object's, and the answer is checked against what a room or a car could be

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A scale with two units is a rate: write it per 1 unit of drawing first.
2. Drawing × rate is real; real ÷ rate is drawing.
3. Area comes from the real dimensions — the rate is used twice, never once.

Slide by Slide

1. **Real-World Scale Drawings.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q4 is the move Notes I makes on a scale: per 1 unit first. *1 retrieves 5.02–03 — area × k^2 (yesterday). 2 retrieves 5.01 — copy ÷ original; k under 1 (earlier this unit). 3 retrieves 4.03 — decimal products; every board today has one (last unit). 4 retrieves MA.6.AR.3.2 — a ratio written as a unit rate; today a scale is read the same way (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 36 cm^2 2. k = 0.6 3. 3.6 4. 45 miles per hour
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the scale and its rewrite.** ¼ in : 1 ft says a quarter inch of paper stands for a foot of room.
6. **Notes.** A room 2 in by 2.5 in at 1 in : 4 ft is 8 ft by 10 ft, so 80 ft^2.
7. **Notes.** A 14-ft wall at 1 in : 4 ft: 14 ÷ 4 = 3.5 in on the paper.
8. **Example 1 — A room's real dimensions and area.** Ask for the scale per 1 inch before anyone multiplies.
9. **Example 1 — Worked.** 1 in : 4 ft; 10 ft by 12 ft; 120 ft^2.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 6 by 16; 96 ft^2.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — The scale is not printed: make it from one pair.** Ask what the scale is before anyone touches the 6 — nobody multiplies until it is written per 1 inch.
13. **Example 2 — Worked.** The pair gives the scale: 18 ÷ 4 = 4.5 ft per inch.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 6.5 ft per inch; 32.5 ft.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. A plan is drawn at $\frac{1}{2}\text{ in} : 1\text{ ft}$. A wall is 4 in long on it. How long is the real wall? → ***8 ft*** *2 — divided (4 × ½) [Notes I: per 1 inch first, then × 2]; 4 — read the scale as 1 : 1*

18–19. Board 2. A plan is drawn at $1\text{ in} : 3\text{ ft}$. A room is 2.5 in long on it. How long is the real room? → ***7.5 ft*** *5.5 — added 3 [7.GR.1.5: the additive error — Math Nation's named misconception]; 0.83 — divided [Notes I: drawing to real is × rate]*

20–21. Board 3. A plan is drawn at $1\text{ in} : 3\text{ ft}$. A real wall is 21 ft long. How long is it on the plan? → ***7 in*** *63 — multiplied [Notes III: real to drawing is a division]; 18 — subtracted 3*

22–23. Board 4. A model car is built at $1 : 50$ and is 8 cm long. How long is the real car? → ***A — 4 m*** *(B) 8 ÷ 50: divided [Notes I: drawing to real is × rate]; (C) 400 with the wrong unit: 400 cm, not 400 m [Notes I: the scale's units are the drawing's]; (D) 8 + 50: the additive error [B1G-M 7.GR.1.5] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. A plan is drawn at $\frac{1}{4}\,\text{in} : 1\,\text{ft}$. A real wall is 15 ft long. How long is it on the plan? → ***3.75 in*** *60 — multiplied by 4 [Notes III: real to drawing is ÷]; 7.5 — divided by 2*

26–27. Board 6. A plan is drawn at $1\,\text{in} : 4\,\text{ft}$. A room is 2 in by 3 in on it. What is the room's real area? → ***B — 96 ft²*** *(A) 6 in² × 4: the drawing's area scaled by the length rate [B1G-M 7.GR.1.5: scaling area like length; Notes II]; (C) 2(8 + 12): the real perimeter, not the area [5.02–03 Notes I]; (D) 8 + 12: the real dimensions added [7.GR.1.5: the additive error — Math Nation's named misconception] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A plane is 36 ft long. One model of it is built at $1:48$, another at $1:72$. Which model is longer, and by how much? → ***The 1 : 48 model, by 3 in (9 in against 6 in)*** *the 1 : 72 model — 'the bigger scale is the bigger model' [Notes III: real ÷ rate — a bigger rate divides more]; 0.75 ft and 0.5 ft with no comparison — stopped early*

30–31. Board 8. Mia, who is 13, enlarges her 9 in by 12 in poster into a mural. The mural's scale is $1\,\text{in} : 2\,\text{ft}$. What are the mural's dimensions? → ***18 ft by 24 ft*** *The 13 (her age) is not needed. 4.5 ft by 6 ft — divided [Notes I: drawing to real is × rate]; 18 in by 24 in — kept the drawing's unit*

32–33. Board 9. On Andrew's drawing a wall is 3.5 in wide and 4 in tall; the real wall is 26.25 ft wide. Find the real height and the real area, and say which step uses the scale. → ***30 ft high; 787.5 ft². The scale (7.5 ft per in) is the step from 4 in to 30 ft.*** *105 ft² — 3.5 × 4 = 14 in², then × 7.5 [B1G-M 7.GR.1.5: scaling area like length]; 'the area step uses the scale' — the scale is between a drawing and a real length, not two real ones*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2 take a drawing length to a real one, the scale changing once; 3 reverses the direction; 4 is a one-unit model scale as a vote; 5 is the quarter-inch scale back to the paper. The pattern breaks at 6, where area is asked and the room has to go to real dimensions before multiplying. 7 compares two models of one object, 8 is the extra number, 9 the B1G-M's wall in writing. The independent set walks the same path: two scales, both directions, a real area, a model in meters, a scale made from one pair.

Misconceptions to Watch

Board 1. 2 — divided (4 × ½) [Notes I: per 1 inch first, then × 2]

Board 2. 5.5 — added 3 [7.GR.1.5: the additive error — Math Nation's named misconception]

Board 3. 63 — multiplied [Notes III: real to drawing is a division]

Board 4. 8 ÷ 50: divided [Notes I: drawing to real is × rate]

Board 5. 60 — multiplied by 4 [Notes III: real to drawing is ÷]

Board 6. 6 in² × 4: the drawing's area scaled by the length rate [B1G-M 7.GR.1.5: scaling area like length

Board 7. the 1 : 72 model — 'the bigger scale is the bigger model' [Notes III: real ÷ rate — a bigger rate divides more]

Board 8. 4.5 ft by 6 ft — divided [Notes I: drawing to real is × rate]

Board 9. 105 ft² — 3.5 × 4 = 14 in², then × 7.5 [B1G-M 7.GR.1.5: scaling area like length]

Materials Whiteboards and markers. A printed floor plan under the document camera for Notes II if the slide's rectangle is not enough. Calculators are allowed on everything and printed on nothing.